LUMINA — AI STUDY ASSISTANT

Grounded AI for Smarter Learning ✨

Project Overview

Lumina is an AI-powered academic assistant built using Retrieval-Augmented Generation (RAG). The system allows users to upload PDF study materials and ask questions directly from the uploaded content. Instead of generating answers from general AI knowledge, Lumina retrieves relevant information from the document first, reducing hallucination and improving factual accuracy.

System Workflow

The pipeline follows a structured process to ensure grounded responses:

Phase	Process Step
Ingestion	PDF Upload → Text Extraction
Indexing	Text Chunking → Embedding Generation → FAISS Vector Index
Retrieval	Semantic Retrieval from Vector Store
Generation	Gemini 2.5 Flash → Grounded Final Answer

Key Features & Technology

- **PDF Processing:** Seamless upload and automated text extraction.
- **Semantic Search:** High-speed retrieval using FAISS Vector Database.
- **Reliability:** Context-aware answer generation with refusal for unknown queries.
- **Interface:** Interactive chatbot experience built for academic focus.
- **Tech Stack:** Python, Google Gemini API, Sentence Transformers, and PyPDF.

Performance Results

- **Accuracy:** Consistent and accurate retrieval from uploaded source material.
- **Verification:** Significant reduction in AI hallucinations.
- **Safety:** Proper handling of queries that fall outside the provided context.
- **Utility:** Supports advanced academic learning and efficient revision cycles.

Sample Interaction

User Query: What is Artificial Intelligence?
Lumina: Artificial Intelligence (AI) is a discipline that enables machines to learn from data and make predictions with minimal human intervention.

User Query: What is Mars?
Lumina: I could not find this in the uploaded study material.

Project Details

- **Author:** Anush Kounain
- **Architecture:** Retrieval-Augmented Generation (RAG)